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Inventor(s)	Fan; Hongyou

Synthesis of ordered nanorod arrays

Abstract

A bi-phased approach between good solvents (or non-polar) and bad solvents (polar) can be used to assemble nanorods into highly ordered monolayers or multilayers of ordered nanorod arrays. These ordered nanorod arrays can display unique optical properties. For example, ordered arrays of CdSe/CdS core/shell nanorods were assembled that display polarized photoluminescence.

Inventors: Fan; Hongyou (Albuquerque, NM)

Applicant: National Technology & Engineering Solutions of Sandia, LLC (Albuquerque, NM)

Family ID: 87162606

Assignee: National Technology & Engineering Solutions of Sandia, LLC (Albuquerque, NM)

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Primary Examiner: Louie; Mandy C

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to nanoparticles and, in particular, to the synthesis of ordered nanorod arrays.

BACKGROUND OF THE INVENTION

(2) Nanorods are unique functional materials. See W. Wei et al., *Angew. Chem. Int. Ed.* 131, 12082 (2019). Different from nanoparticle with spherical morphology, nanorods are one-dimensional and

have anisotropic morphology that often show unique properties. The ability to assemble nanorods into ordered arrays to exhibit anisotropic function is crucial for their nano-electronic and optical applications. Current approaches are essentially limited to aqueous processes using hydrophilic nanorods. See N. R. Jana et al., *J. Mater. Chem.* 12, 2909 (2002). These aqueous processes rely on evaporation induced drying or assembly at liquid-vapor interface, which leads to ill-defined morphology of the ordered arrays. Additionally, drying stress often crack the final arrays.

SUMMARY OF THE INVENTION

(3) The present invention is directed to a method to synthesize ordered nanorod arrays, comprising providing a colloidal suspension of nanorods having a hydrophobic surface in a non-polar solvent, and counter-diffusing the non-polar solvent and a polar anti-solvent into each other, thereby slowly precipitating ordered nanorod arrays from the colloidal suspension. The counter-diffusing can comprise adding the polar anti-solvent on top of the colloidal suspension and allowing the non-polar solvent to counter-diffuse into each other. The nanorods can have an aspect ratio equal to or greater than about 3 (i.e., $\pm 10\%$) and can be assembled into ordered nanorod arrays comprising one or more hexagonal monolayers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description will refer to the following drawings, wherein like elements are referred to by like numbers.

(2) FIG. 1 is a schematic illustration of the synthesis of an ordered nanorod array.

(3) FIGS. 2A-2D are transmission electron microscope (TEM) images of CdSe/CdS core/shell nanoparticles and nanorods. FIG. 2A is a TEM image of spherical nanoparticles (SpNP). FIG. 2B is a TEM image of short nanorods (SNR). FIG. 2C is a TEM image of medium nanorods (MNR). FIG. 2D is a TEM image of long nanorods (LNR). Scale bars are 50 μm .

(4) FIGS. 3A-3D are scanning electron microscope (SEM) images of the self-assembled superstructures of CdSe/CdS core/shell nanoparticles and nanorods. FIG. 3A is an SEM image of 3D supercrystals of SpNP. FIG. 3B is an SEM image of multilayered flakes of SNR. FIG. 3C is an SEM image of a 2D monolayer of MNR. FIG. 3D is an SEM image of a monolayer of LNR. Scale bars are 50 μm . FIGS. 3E-3H are corresponding high resolution images of edges of the superstructures above showing hexagonal packing of the constituent particles. Scale bars are 50 nm.

(5) FIG. 4 is a graph of small-angle X-ray scattering (SAXS) patterns of the self-assembled superstructures of CdSe/CdS core/shell nanoparticles and nanorods of different aspect ratios. The Miller indices of the 3D fcc superlattice of SpNP and the 2D hexagonal monolayer superlattice of the LNR samples are labeled at the bottom and top respectively. The inset schematics illustrate the three different morphologies.

(6) FIG. 5A is a TEM image showing vertically aligned NRs in the monolayer of LNRs shown in FIGS. 3D and 3H. Scale bar is 10 nm. FIG. 5B is a wide-angle X-ray scattering (WAXS) pattern showing a strong (002) peak confirming the uniform vertical alignment of the nanorods on the substrate. FIG. 5C is a schematic of an individual nanorod showing the wurtzite crystal structure, the (002) planes of this nanorod are parallel with the underlying substrate, corresponding to the WAXS pattern.

(7) FIGS. 6A-6D show photoluminescence (PL) polarization of LNR thin films of two different configurations, flakes and reference. FIG. 6A is a schematic illustration of the PL measurement. The emission intensity of two orthogonal polarization directions vertical (v) and horizontal (h) to the substrate. FIGS. 6B and 6C are graphs of the integrated PL emission intensity at varying detection angle θ of the hexagonal monolayer flake and reference sample, respectively. The insets

demonstrate the configuration of nanorods on glass substrates in each sample. The drop of intensity above $\theta=60^\circ$ is caused by partial lens blockage by the substrate. FIG. 6D is a graph of intensity ratio between vertical and horizontal polarization for the two configurations. Strong vertically polarized emission was observed from LNR flakes due to the “standing up” configuration of nanorods.

DETAILED DESCRIPTION OF THE INVENTION

(8) The present invention is directed to the synthesis of highly ordered arrays (e.g., flakes, monolayers, superdisks, etc.) of nanorods that display unique optical properties. This invention uses a bi-phased approach between good solvents (or non-polar) and bad solvents (polar) to assemble nanorods into highly ordered monolayers or multilayers of ordered nanorod arrays. A related bi-phased approach has been used to assemble isotropic spherical nanoparticle arrays, but has not been used to assemble one-dimensional nanomaterials. See U.S. Pat. No. 10,947,116 to H. Fan and K. Bian. Through guidance of theoretical calculation, this invention allows the controlled synthesis of ordered arrays.

(9) The method to synthesize ordered nanorod arrays comprises providing a colloidal suspension of nanorods having a hydrophobic surface in a non-polar solvent, and counter-diffusing the non-polar solvent and a polar anti-solvent into each other, thereby forming a solvent/anti-solvent mixture and slowly precipitating the nanorods into an ordered nanorod array. The ordered nanorod arrays can comprise semiconductor, metal, or magnetic nanorods, such as semiconducting CdS, CdSe, ZnS, and ZnSe, metallic gold and silver, and magnetic Fe₃O₄ and FePt. See L. Carbone et al., *Nano Lett.* 7(10), 2942 (2007); N. R. Jana et al., *J. Phys. Chem. B* 105, 4065 (2001); and M. Chen et al., *J. Am. Chem. Soc.* 129(20), 6348 (2007). The nanorods can further comprise rigid hydrophobic organic macromolecules such as porphyrins and phthalocyanine. For example, the hydrophobic surface can comprise an alkyl, such as alkyl thiol, alkyl amine, alkyl acid, or alkyl phosphine. For example, the non-polar solvent can comprise toluene, hexane, benzene, or xylene. For example, the polar anti-solvent can comprise isopropanol, ethanol, methanol, or tetrahydrofuran. The size and quality of the ordered nanorod arrays depend on the initial nanorod concentration of diffusion speed. The ordered nanorod arrays can be greater than one hundred microns in lateral dimension.

Synthesis of CdSe Seeds

(10) As an example of the invention, CdSe/CdS nanorods (NRs) were synthesized using a seeded method according to Carbone. See L. Carbone et al., *Nano Lett.* 7(10), 2942 (2007). To make the CdSe nanoparticles seeds, 240 mg CdO, 1.26 g octadecylphosphonic acid (ODPA) and 12 g trioctylphosphine oxide (TOPO) were added in a 100 mL three-neck flask. The mixture was melted and heated to 150° C. under vacuum for 1 hour while being stirred to degas the mixture. The resulting CdO suspension was heated to 310° C. under argon gas to form a clear solution into which 6 mL trioctylphosphine (TOP) was injected. The solution was then heated to 370° C. and 1.7 mL of TOP:Se (1.7 M) complex was quickly injected. The solution immediately became orange/red in color indicating formation of CdSe nanoparticles. After reaction for 1 minute, the heat mantle was removed and the flask was cooled down in air. The raw product was separated and cleaned by three cycles of precipitation with acetone and toluene. The purified nanoparticles were dissolved in TOP to form a stock CdSe seed solution. The stock solution was characterized to contain 2.9 nm CdSe nanoparticles with a concentration of 0.4 mM.

Synthesis of CdSe/CdS Core/Shell Nanorods

(11) To synthesize the CdSe/CdS core/shell nanorods, in a 50 mL three-neck flask, 0.086 g CdO as Cd precursor was mixed with 3 g TOPO, 0.29 g ODPA and 80 mg hexaphosphoric acid (HPA) and heated to 150° C. under vacuum for one hour to degas the mixture and then to 350° C. under Ar to form a clear Cd precursor solution. Meanwhile, in an Ar glove box, a sulfur precursor solution was prepared by dissolving 0.12 g sulfur powder in 2 mL TOP at 110° C. then mixing the sulfur precursor solution with 0.2 mL CdSe seed solution. The seeded sulfur precursor solution was then

quickly injected into the Cd precursor solution. The reaction proceeded for 6 min before the heating mantle was removed to allow the reactor to cool down in air. The raw product was separated and cleaned by three cycles of precipitation with acetone and toluene. The purified CdSe/CdS core/shell nanorods having a hydrophobic trioctylphosphine surface were redispersed in toluene. Nanorods of different sizes were achieved by changing Cd and S precursor concentrations correspondingly.

Synthesis of Spherical CdSe/CdS Core/Shell Nanoparticles

(12) Spherical CdSe/CdS core/shell nanoparticles were synthesized using method reported by Chen. See O. Chen et al., *Nat. Mater.* 12(5), 445 (2013). Briefly, in a 100 mL flask, 0.5 mL CdSe seed solution was added into 6 mL octadecene (ODE) and 6 mL oleylamine. The mixture was degassed at 120° C. for 1 hr then heated to 310° C. under Ar. Using a syringe pump, 6 mL of ODE solution of octanethiol (0.13M) and cadmium oleate (0.13M) were slowly (6 mL/hr) injected simultaneously into the seed solution as S and Cd precursors, respectively. Then 2 mL oleic acid was injected to further stabilize the nanoparticles before cooling the reactor in air. The raw product was separated and cleaned by three cycles of precipitation with acetone and toluene. The purified CdSe/CdS core/shell nanoparticles having a hydrophobic oleylamine surface were redispersed in toluene.

Growth of Ordered Nanorod Arrays and Thin Film Deposition

(13) The ordered nanotube arrays were grown by a counter-diffusion method that has previously been used to grow supercrystals from nanoparticles. See U.S. Pat. No. 10,947,116 to H. Fan and K. Bian, which is incorporated herein by reference. As shown in FIG. 1, a vertically positioned test tube was filled with solutions of the synthesized nanorods or nanoparticles suspended in the non-polar solvent toluene. The anti-solvent isopropanol (IPA) was then carefully added on top of the toluene suspension with a volumetric ratio of IPA:toluene=4:1. A liquid-liquid interface between the two solvents was formed. The total height of the liquid was about 6 cm. The tubes was left undisturbed as the counter-diffusion of the two solvents into each other proceeded, as indicated by a blurred interface. As the anti-solvent IPA concentration increased in the toluene solvent phase, the nanorods or nanoparticles became over saturated and slowly precipitated. The growth was considered complete when the interface disappeared and the mixture became homogenous and colorless. The product ordered nanorod arrays or nanoparticle supercrystals were collected by removing the liquid, rinsed, and then stored in ethanol. To deposit a monolayer film of the ordered nanorod arrays on a substrate, their solutions were added dropwise on the surface of water in a Petri dish until a continuous film was formed. Then a substrate, either glass or Si wafer, was placed on the water surface and withdrawn to pick up the ordered array film, which was then dried and characterized.

(14) Four different sizes of nanoparticles/nanorods were synthesized, as shown in in FIGS. 2A-2D and Table 1. The average particle size was determined by statistics from multiple TEM images.

(15) TABLE-US-00001 TABLE 1 List of CdSe/CdS core/shell nanorods and nanoparticles.

Diameter (nm)	Aspect Ratio	d Length (nm)	k = d/L	Sample Designation
6.8 nm ± 8.8%	N/A	1.0	1.0	Spherical SpNP
5.4 nm ± 8.5%	16.1	3.0	7.8%	Short nanorod SNR
5.2 nm ± 8.7%	27.3	5.3	6.7%	Medium nanorod MNR
6.1 nm ± 8.5%	33.7	5.5	6.8%	Long nanorod LNR

(16) The nanoparticles self-assembled into a faceted 3D supercrystal, as shown in the SEM images in FIGS. 3A and 3E. The nanorods self-assembled into different hexagonal ordered arrays, including monolayer and multilayer flakes depending on aspect ratio, as shown in FIGS. 3B-3D and 3F-3H. As shown in FIG. 4, SAXS patterns revealed the superlattice structures. The structure parameters summarized in Table 2. For SpNP, face-center cubic (fcc) supercrystals were formed. Due to the spherical interactions, no orientational order was expected. For SNR, hexagonal multilayered flakes as a transition state were formed with poor ordering. Orientational order was not well established due to the low aspect ratio of the short nanorods. For MNR, hexagonal monolayers were formed with all the nanorods standing up vertically to the monolayer plane. For

LNR, hexagonal monolayers were formed with all the nanorods standing up vertically to the monolayer plane.

(17) TABLE-US-00002 TABLE 2 Structure parameters of the superlattices Nearest Neighbor Separation distance d.sub.NN d.sub.NN – d Sample Superlattice Lattice Parameters (nm) (nm) SpNP fee a = b = c = 13.8 nm 9.7 2.9 supercrystal $\alpha = \beta = \gamma = 90^\circ$ SNR Hex. N/A 6.3 0.9 Multilayer MNR Hex. a = b = 6.3 nm 6.3 1.1 Monolayer $\alpha = 120^\circ$ LNR Hex. a = b = 7.5 nm 7.5 1.4 Monolayer $\alpha = 120^\circ$

(18) A model to predict superstructure morphology was developed to correlate with the aspect ratio, assuming a self-assembled flake made of piled circular monolayers of nanorods. Nanorods were characterized as cylinders with flat head and bottoms.

(19) The tip surface area is

$$A_{\text{sub.tip}} = \pi d_{\text{sup}}^2 / 4 \quad (1)$$

where d is the diameter of an individual nanorod. The side surface area is

$$A_{\text{sub.side}} = \pi d l \quad (2)$$

where l is the length of an individual nanorod. The number of nanorods in one monolayer is

$$N/m \quad (3)$$

where N is the total number of nanorods in a self-assembled flake and m is the number of hexagonal monolayers in a flake, i.e., $m \geq 1$. Therefore, the total surface energy of exposed tips in a flake (side+bottom, $\times 2$) is:

$$(20) \quad E_{\text{tip}} \times 2 \times \frac{N}{m} \times \frac{d^2}{4} = \frac{E_{\text{tip}} N d^2}{2m} \quad (4)$$

The radius R of a monolayer is

$$(21) \quad R = \sqrt{\frac{N A_{\text{tip}}}{0.907 m}} = 0.525 d \sqrt{\frac{N}{m}} \quad (5)$$

where 0.907 is packing density of a 2D hexagonal lattice. The number of nanorods expose on the side in a monolayer:

$$2\pi R/d \quad (6)$$

The total surface energy of exposed tips in a flake (each nanorod exposes half of its side area, assuming $R \gg r$)

$$(22) \quad E_{\text{side}} \times \frac{2}{d} R \times \frac{1}{2} \times A_{\text{side}} = 0.525 k E_{\text{side}}^2 d^2 \sqrt{\frac{N}{m}} \quad (7)$$

where $E_{\text{sub.side}}$ is the surface energy of side and $E_{\text{sub.tip}}$ is the surface energy of a tip, and k is the aspect ratio for the nanorod. The total surface energy of the flake, assuming $E_{\text{sub.side}} \cong E_{\text{sub.tip}} = E_{\text{sub.surf}}$ for simplicity, is:

$$(23) \quad E_{\text{total}} = \frac{1}{2} d^2 E_{\text{surf}} \left(\frac{N}{m} + 1.05 k \sqrt{mN} \right) \quad (8)$$

The optimized structure should have $m_{\text{sub.opt}}$ that gives lowest $E_{\text{sub.total}}$:

$$(24) \quad \frac{\partial E_{\text{total}}}{\partial m} = \frac{1}{2} d^2 E_{\text{surf}} \left(0.525 k \sqrt{\frac{N}{m}} - \frac{N}{m^2} \right) = 0 \quad (9) \quad m_{\text{opt}} = 0.72 \sqrt{\frac{N}{k^2}} \quad (10)$$

Now, consider the following special cases: 1. Thin disk, $k=0$,

(25) $E_{\text{total}} = \frac{1}{2} d^2 E_{\text{surf}} \frac{N}{m}$, $m=N$ gives minimum energy—coins piled in one column 2. “Sphere”, $k=1$,

(26) $m_{\text{opt}} = 0.72 \sqrt[3]{N}$, 3D supercrystal with growth in all three dimensions 3. Needles, $k=\infty$, $E_{\text{sub.total}} = 0.525 \pi d_{\text{sup}}^2 d_{\text{sup}}^2 E_{\text{sub.surf}} k \sqrt{mN}$, $m=1$ gives minimum energy for monolayer flakes. 4. Short nanorods ($k < 0.61 \sqrt[3]{N}$) will lead to a

(27) $m_{\text{opt}} = 0.72 \sqrt[3]{N} > 1$, i.e. multilayers. 5. Long nanorods ($k > 0.61 \sqrt[3]{N}$) will lead to a $m_{\text{sub.opt}}=1$, i.e. monolayer.

(28) According to Table 2, nanorods have greater interdigitation (3 nm vs ~1 nm) on the sides compared with spherical nanoparticles which resemble the tips of the nanorods. Therefore, a higher energy is expected at the tips of the nanorods than at the side. $E_{\text{sub.tip}} > E_{\text{sub.side}}$. This can shift the conclusion of the model quantitatively but not qualitatively. It increases the tendency toward

the monolayer configuration. This can explain the difference between model results and experimentally observed results. Ignoring the curvature of the tips of the nanorods also leads to deviations.

(29) This simplified model assumes the self-assembled structures can always find the globally low energy configuration. This is not the case in the actual assembly process. Once a nanorod is attached to a flake by oversaturation-driven precipitation, it becomes immobile. Therefore, it cannot “feel” the entire flake but only the nearby neighbors. This is why simply plugging in the experimental numbers of nanorods in a flake into the model predicts a $m_{sub,opt}$ much higher than the actual results. However, the size of initial nuclei of these flakes is expected to be comparable with N , the total number of nanorods in the flake. So, based on the observed configurations for different aspect ratios k summarized in Table 1, it can be estimated that

$(k_{sub,SNR}/0.61)^{sup.2} < N < (k_{sub,MNR}/0.61)^{sup.2}$. That is to say, a nanorod can only “feel” a few tens of its neighbors.

(30) The structure of a monolayer sample of a long nanorod is shown in FIG. 5. The TEM in FIG. 5A shows the nanorods in the “stand up” configuration in a monolayer. This was confirmed by WAXS in FIG. 5B.

(31) Polarized photoluminescence measurements were performed by a custom-built optical system, as shown in FIG. 6A. A 532 nm laser was focused to excite the sample on a glass slide. The incident angle was 15° to avoid specular reflection being detected. The emitted light was converged by an objective lens into an optic fiber which guided the emission to an optical spectrometer. Before the objective lens, the emission light passed a polarizer set either parallel or perpendicular to the sample film. The emission spectra were collected at reflection angles θ of 10° 20° . . . 80° . A bare glass slide was used as reference. The emission intensity was calculated by integrating the area under the luminescence peak.

(32) Optical experiments on the self-assembled long nanorod monolayer flakes (superstructure, ordered) drop cast on a glass slide from an ethanol suspension were compared with the reference sample (random, disordered) formed by a long nanorod solution in toluene drop cast on a glass slide. Photoluminescence polarization measurements, shown in FIG. 6B, indicate preferential vertically polarized emission of the flake sample deposited on the glass slide. As shown in FIG. 6C, the reference sample with nanorods lying on the substrate with random orientation showed slight horizontal polarity at grazing angle. The drop of intensity above $\theta=60^\circ$ is caused by partial lens blockage by the substrate. Comparing with reference sample, the emission from the monolayer flakes displayed high polarity vertical to the substrate, as shown in FIG. 6D. This is a direct consequence of the vertically aligned nanorods in the flake.

(33) The present invention has been described as synthesis of ordered nanorod arrays. It will be understood that the above description is merely illustrative of the applications of the principles of the present invention, the scope of which is to be determined by the claims viewed in light of the specification. Other variants and modifications of the invention will be apparent to those of skill in the art.

Claims

1. A method to synthesize ordered nanorod arrays, comprising: providing a colloidal suspension of nanorods having a hydrophobic organic surface in a non-polar solvent, and counter-diffusing the non-polar solvent and a polar anti-solvent into each other to form a solvent/anti-solvent mixture, thereby slowly precipitating the nanorods having the hydrophobic organic surface from the colloidal suspension to form ordered nanorod arrays.
2. The method of claim 1, wherein the counter-diffusing comprises adding the polar anti-solvent on top of the colloidal suspension and allowing the non-polar solvent and the polar anti-solvent to counter-diffuse into each other.

3. The method of claim 1, wherein the nanorods have an aspect ratio equal to or greater than about 3.
 4. The method of claim 1, wherein the ordered nanorod arrays comprise one or more hexagonal monolayers.
 5. The method of claim 1, wherein the ordered nanorod arrays comprise semiconductor, metal, or magnetic nanorods.
 6. The method of claim 5, wherein the semiconductor nanorods comprise CdS, CdSe, ZnS, ZnSe, or combinations thereof.
 7. The method of claim 6, wherein the semiconductor nanorods comprise CdSe/CdS core/shell nanorods.
 8. The method of claim 5, wherein the metal nanorods comprise gold or silver.
 9. The method of claim 5, wherein the magnetic nanorods comprise Fe₃O₄ or FePt.
 10. The method of claim 1, wherein the hydrophobic organic surface comprises an alkyl thiol, alkyl amine, alkyl acid, or alkyl phosphine.
 11. The method of claim 1, wherein the non-polar solvent comprises toluene, hexane, benzene, or xylene.
 12. The method of claim 1, wherein the polar solvent comprises isopropanol, ethanol, methanol, or tetrahydrofuran.
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